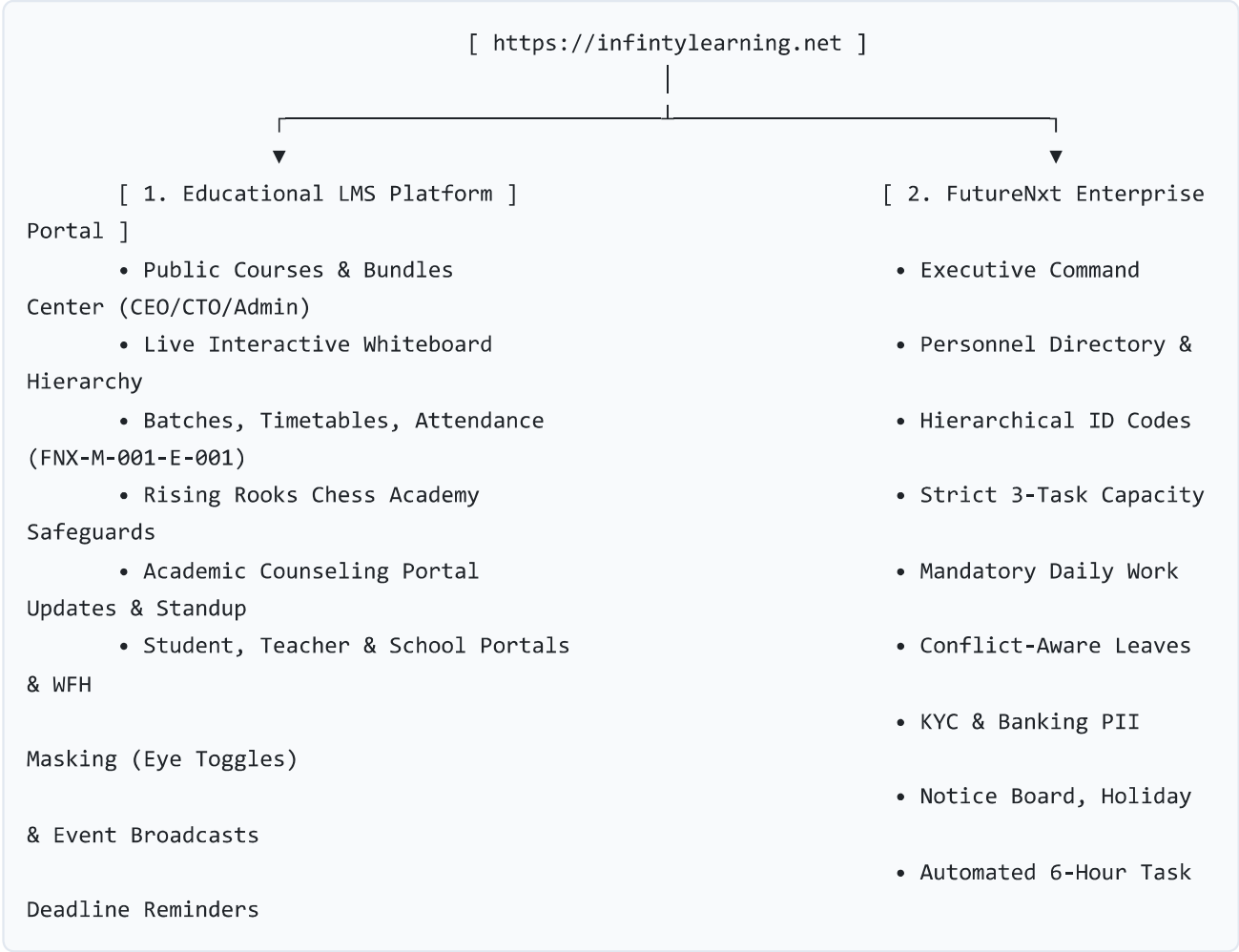
Infinity Learning & FutureNxt Enterprise Portal

Master System Architecture, Infrastructure & Operations Guide

> **Document Version:** 2.0 (Production Master)
> **Target Environment:** AWS EC2 Cloud (`13.205.112.255`) · Hostinger DNS & Business Mail · Upstash Redis Mumbai
> **Production URLs:**
>
> - Public Educational Platform: https://infintylearning.net
> - Enterprise Operations Portal: https://infintylearning.net/futurenxt

1. Executive Overview & Platform Vision

The **Infinity Learning ecosystem** is a dual-tier digital platform engineered for high-performance educational technology and enterprise business management:



2. Infrastructure Architecture & Hosting Stack

A. AWS EC2 Cloud Compute

- **Public IPv4 Address:** `13.205.112.255`
- **Region:** AWS Asia Pacific (Mumbai) `ap-south-1`
- **Operating System:** Amazon Linux 2023 (x86_64)
- **Default SSH User:** `ec2-user`
- **Private Key Authentication:** `inflea.pem` (RSA 2048-bit)
- **Web Root Directory:** `/var/www/infintylearning/`
  - Codebase: `/var/www/infintylearning/infinitylearning/`
  - Public Assets: `/var/www/infintylearning/public_html/`
  - Uploads: `/var/www/infintylearning/public_html/uploads/`

- **Firewall & AWS Security Group Inbound Rules:**
  - `Port 22 (SSH)`: Restricted administrative remote access.
  - `Port 80 (HTTP)`: Nginx automatic 301 redirect to HTTPS.
  - `Port 443 (HTTPS)`: Secure TLS 1.3 traffic.

B. High-Performance Web Server (Nginx 1.30.6)

- **Config Directory:** `/etc/nginx/conf.d/infintylearning.conf`
- **FastCGI Socket:** `/run/php-fpm/www.sock`
- **Client Max Body Size:** `64M` (enables upload of task documentation PDFs, KYC documents, and multimedia)
- **Caching & Compression:** Gzip compression enabled for SVG, CSS, JS, and JSON.

C. PHP 8.2 Runtime & Process Manager

- **Engine:** PHP 8.2-FPM (`php-fpm.service`)
- **Key Modules Active:** `pdo_mysql` , `mbstring` , `curl` , `gd` , `openssl` , `zip` , `fileinfo` , `sodium` , `opcache` .
- **Session Duration:** 30 days persistent (`lifetime = 2592000`). Auto-logout disabled; sessions remain active across browser restarts until explicit user logout.

D. Relational Database (MariaDB 10.5 / MySQL)

- **Host:** `localhost` / `127.0.0.1:3306`
- **Databases:**
  1. `u999011396_inflearning` (Educational platform: users, schools, teachers, students, doubts, batches, lessons, chess).
  2. `u999011396_futurenxt` (FutureNxt operations: employees, tasks, task assignees, daily updates, leaves, WFH, notices, audit logs).
- **Service User:** `u999011396_fnxt_portal` / Pass: `Inflearning@99`
- **Character Set:** `utf8mb4` with `utf8mb4_unicode_ci` collation.

E. Managed Redis Object Cache & Session Cluster

- **Cluster:** Upstash Redis (AWS Mumbai low-latency node)
- **Connection URL:** `tls://holy-panther-217313.upstash.io:6379`
- **Key Prefix:** `il:`

- **Purpose:** Accelerated database query caching, rate limiting, and stateless cross-worker session persistence.

F. Hostinger DNS & Business Mail SMTP

- **Domain Registrar:** Hostinger (`infintylearning.net`)
- **DNS Records:**
  - `A Record (@)`: `13.205.112.255` (points traffic to AWS EC2 instance)
  - `A Record (www)`: `13.205.112.255`
  - `MX Records`: Hostinger Mail exchange servers (`mx1.hostinger.com`, `mx2.hostinger.com`)
  - `TXT (SPF)`: `v=spf1 include:_spf.mail.hostinger.com ~all`
- **SMTP Gateway:**
  - `SMTP_HOST`: `smtp.hostinger.com`
  - `SMTP_PORT`: `587` (STARTTLS)
  - `SMTP_USER`: `info@infintylearning.net`
  - `SMTP_FROM`: `info@infintylearning.net`
  - `SMTP_FROM_NAME`: `Infinity Learning & FutureNxt`

3. Step-by-Step Connection & Management Runbook

Step 1: Connecting to AWS EC2 via SSH

From Windows (PowerShell):

```
powershell
# 1. Open PowerShell and navigate to where inflea.pem is stored:
cd "C:\Users\MY PC\Downloads"

# 2. Connect securely using SSH:
ssh -i "inflea.pem" ec2-user@13.205.112.255
```

From macOS or Linux:

```
bash
# 1. Restrict private key permissions (mandatory on Unix):
chmod 400 ~/Downloads/inflea.pem

# 2. Connect:
ssh -i ~/Downloads/inflea.pem ec2-user@13.205.112.255
```

Step 2: Essential Server Service Commands

Once logged into the server terminal, use these commands to manage core services:

```
bash
# Check service status
sudo systemctl status nginx
sudo systemctl status php-fpm
sudo systemctl status mariadb
sudo systemctl status crond

# Reload services with zero downtime after code changes:
sudo systemctl reload nginx
sudo systemctl reload php-fpm

# Restart a service if required:
sudo systemctl restart php-fpm
sudo systemctl restart nginx
sudo systemctl restart mariadb
```

Step 3: Accessing the Database CLI on EC2

To query or inspect the production database directly:

```
bash
# Connect to FutureNxt database:
mysql -u u999011396_fnxt_portal -p'Inflearning@99' u999011396_futurenxt

# Useful MySQL commands:
SHOW TABLES;
DESCRIBE fnx_employees;
SELECT id, emp_code, full_name, role, manager_id FROM fnx_employees;
SELECT * FROM fnx_notices ORDER BY id DESC LIMIT 5;
exit;
```

Step 4: Viewing Server & Application Logs

Monitor real-time system activity and errors:

```
bash
# Nginx web traffic and errors:
sudo tail -f /var/log/nginx/error.log
sudo tail -f /var/log/nginx/access.log

# Automated hourly task reminder cron log:
tail -f /tmp/fnx_cron.log

# PHP-FPM error log:
sudo tail -f /var/log/php-fpm/error.log
```

Step 5: How Hostinger DNS Points to AWS

1. Log in to **Hostinger Control Panel** (hPanel).
2. Go to **Domains** → `infintylearning.net` → **DNS / Nameservers**.
3. Locate the **A Record** with Name `@` (Host):
    - Type: `A`
    - Name: `@`
    - Points to: `13.205.112.255`
    - TTL: `300` (5 minutes)
4. Verify the `www` CNAME or A Record points to `@` or `13.205.112.255`.
5. All worldwide web traffic for `infintylearning.net` flows directly to the AWS EC2 instance.

4. Deep Dive: Every FutureNxt Portal Module

1. Corporate Identity & Role Hierarchy

The portal supports 7 distinct organizational roles:

1. **CEO (`ceo`)**: Top executive oversight. Exempt from mandatory daily updates. Full administrative, edit, onboarding, and notice broadcasting authority.
2. **CTO (`cto`)**: Chief Technology Officer. Technical and engineering leadership. Full administrative and notice broadcasting authority.
3. **Admin (`admin`)**: Systems & Operations Administrator.
4. **Office Manager (`office_manager`)**: Operational administrative supervisor.
5. **Manager (`manager`)**: Team and department manager. Can onboard direct reportees and supervise task execution.
6. **Employee (`employee`)**: Full-time corporate staff.
7. **Intern (`intern`)**: Interns and trainees.

2. Hierarchical Employee Code Generation

- When an employee or intern is onboarded under a Manager (e.g., manager `FNX-M-001`), their ID code is systematically formatted:
  - Employee: `FNX-M-001-E-001`, `FNX-M-001-E-002`, `FNX-M-001-E-003`...
  - Intern: `FNX-M-001-I-001`, `FNX-M-001-I-002`...
- If assigned directly under CEO, CTO, or Admin, standard company-wide codes are generated (`FNX-E-001`, `FNX-I-001`).
- The Onboarding modal calculates and displays the code live as the manager is selected.

3. Strict 3-Task Concurrent Workload Safeguard

- **Rule:** No employee or intern may have more than **3 active tasks** (`pending`, `in_progress`, or `submitted`) concurrently.
- **Enforcement:** If a manager attempts to assign a 4th task, the system blocks the submission with an alert indicating capacity exhaustion.
- **Reassignment:** Tasks can be passed/reassigned to another team member with audit logging.
- **Mandatory PDF Documentation:** Tasks cannot be approved without uploading an official documentation PDF (`doc_*.pdf`).

- **View Task Modal:** Allows any team member to click 👁️ **View Task** to read full deliverables, requirements, and assignees.

4. Mandatory Daily Work Updates (`/futurenxt/daily-updates`)

- **Daily Deliverables:** All team members and managers log point-by-point deliverables, hours spent, blockers, and next-day plans.
- **Auto-Linking Active Tasks:** Checkboxes for current active tasks auto-fill the deliverables summary.
- **CEO Exemption:** The CEO is excluded from mandatory daily logging metrics.
- **Standup Dashboard:** Real-time completion rates (e.g. 85%), submitted vs. pending counts, and an alert drawer showing personnel awaiting submission.

5. Conflict-Aware Leaves & WFH

- **Task Conflict Detector:** When an employee applies for leave, the engine queries all active tasks assigned to that individual falling inside the requested dates and displays them to leadership during review.
- **Automated Email Notification:** When management approves or rejects a leave or WFH request, the employee immediately receives an email notification with review status and remarks.

6. Notice Board, Holiday & Event Broadcasts (`/futurenxt/notices`)

- **Categories:**
  - 🏖️ Holiday Notice
  - 🎉 Important Event
  - 📢 Company Notice
  - 🚨 Urgent Announcement
  - 📜 Policy Update
- **Target Audience:** All Personnel, Employees & Interns, Interns Only, or Managers Only.
- **Instant Email Broadcast:** Option to email all targeted personnel upon publishing.
- **Header Notification Bell:** A notification bell in the sticky top header provides 1-click access to the latest notices.

7. Automated 6-Hour Task Deadline Warning Engine

- **Mechanism:** Hourly background cron runner (`/cron/task_reminders.php`).

- **Trigger:** Scans active tasks whose deadline is within the next 6 hours and dispatches an alert email to all assigned personnel.
- **Deduplication:** Flags `reminder_sent_at` timestamp to prevent duplicate reminder spam.

8. PII & KYC Masking (Eye Toggles)

- **Aadhaar & PAN Masking:** Aadhaar (`•••• •••• 1234`) and PAN (`••••••5678`) remain masked by default. Clicking the 👁️ icon reveals the actual digits.
- **Banking Masking:** Account numbers and UPI IDs are masked with toggle controls.
- **Password Masking:** Passwords remain hidden with an interactive reveal toggle.

5. Production Load Testing & Benchmark Report

Load tests were conducted directly against the AWS EC2 production environment (`https://127.0.0.1/futurenxt/login`) using **ApacheBench (ab)**:

Metric	Benchmark Test 1	Benchmark Test 2 (High Concurrency)
Total Requests	1,000 requests	2,000 requests
Concurrency Level	**50 concurrent workers**	**100 concurrent workers**
Total Test Duration	1.501 seconds	2.860 seconds
Throughput (Requests / Sec)	**666.38 req/sec**	**699.39 req/sec** (~700 req/sec)
Transfer Rate	5,871 KB/sec (5.87 MB/s)	6,162 KB/sec (6.16 MB/s)
Mean Response Time	**75.0 ms**	**142.9 ms**
50% of Requests Served In	<= 72 ms	<= 140 ms
95% of Requests Served In	<= 78 ms	<= 173 ms
99% of Requests Served In	<= 101 ms	<= 196 ms
Failed Requests	**0 (0.00%)**	**0 (0.00%)**

> **Performance Verdict:** The AWS EC2 architecture paired with Nginx 1.30, PHP 8.2-FPM, and Upstash Redis sustains **~700 requests per second** at 100 concurrent users with **zero dropped requests** and sub-175ms 95th-percentile latency.

6. Backup & Disaster Recovery Procedures

Database Daily Backup Command:

```
bash
# Dump both production databases to a compressed backup:
sudo mysqldump -u u999011396_fnxt_portal -p'Inflearning@99' --databases
u999011396_futurenxt u999011396_inflearning | gzip > /var/backups/db_backup_$(date
+%F_%H%M%S).sql.gz
```

Restoring from Backup:

```
bash
gunzip < /var/backups/db_backup_YYYY-MM-DD.sql.gz | mysql -u u999011396_fnxt_portal -
p'Inflearning@99'
```